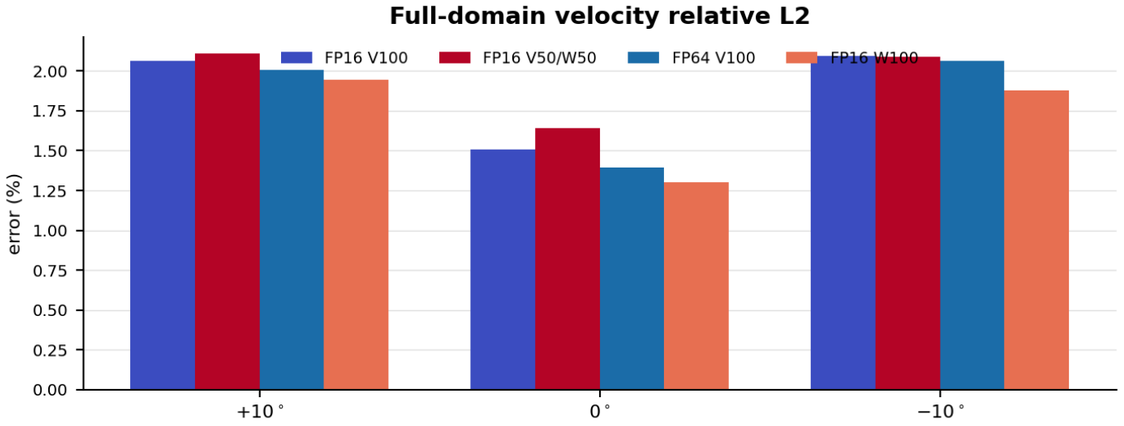
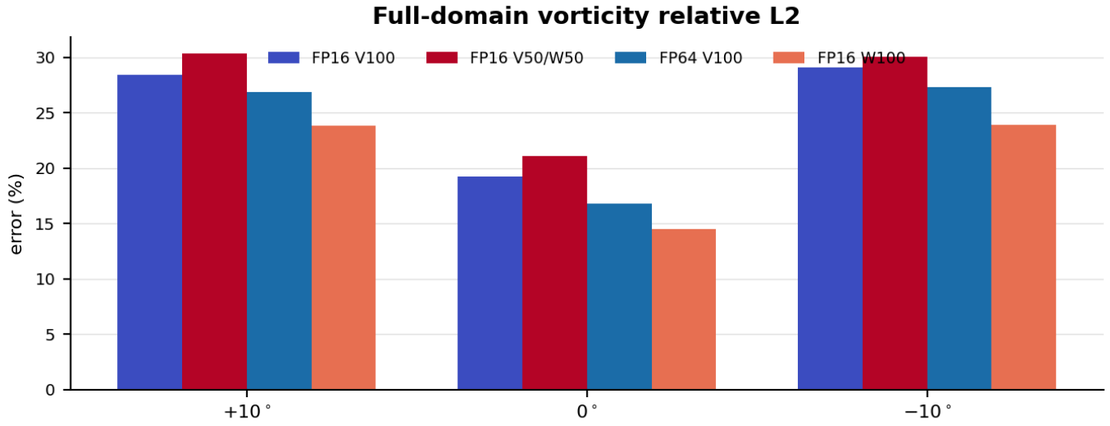
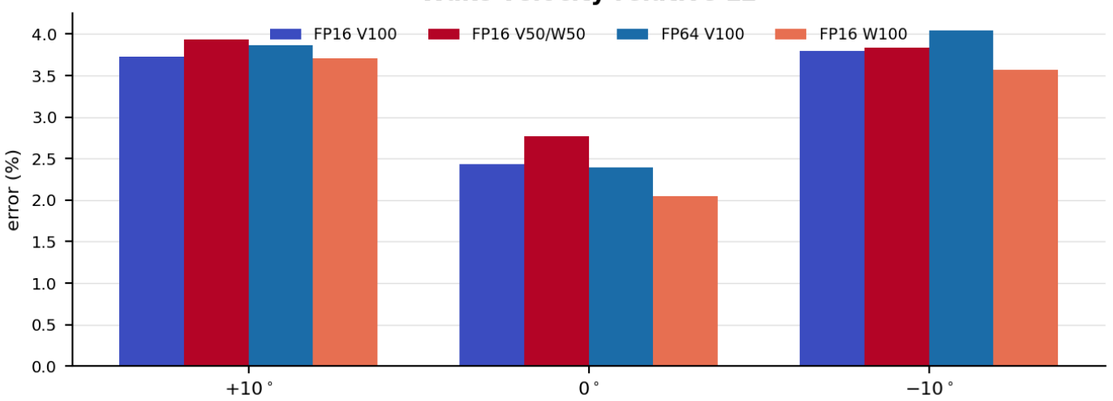
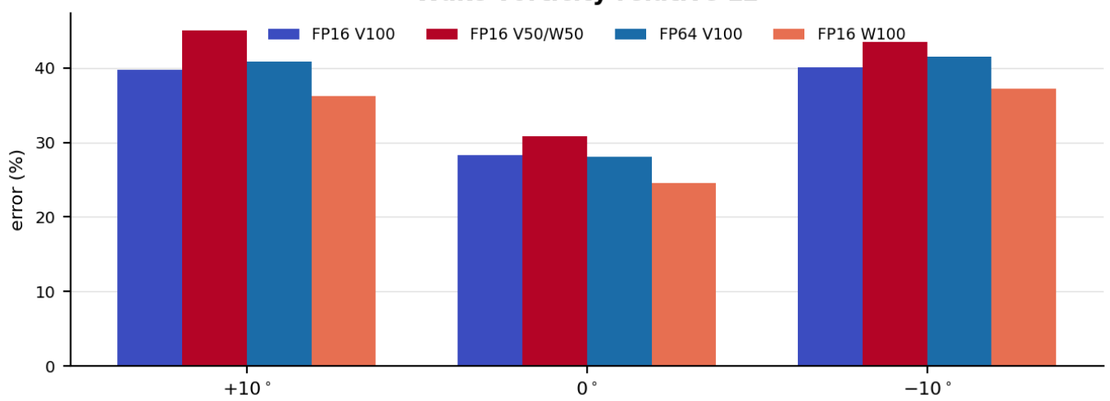
Airfoil four-model, three-frame L2 comparison (vorticity epoch 5000)

k=1024 | Re=1500 | frame 20 | velocity-L^2 coefficient fit | identical GT fields







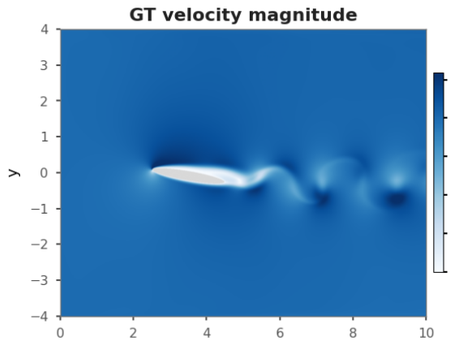


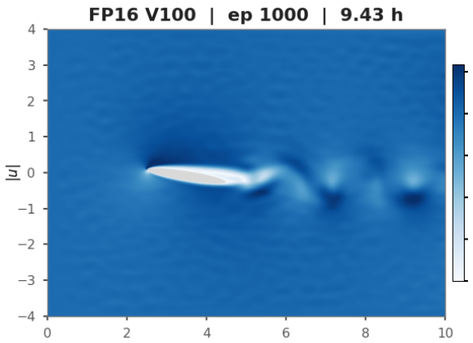
Model	Weights V/W	Evaluated epoch	Time to checkpoint	Mean L2(u)	Mean L2(omega)	Mean wake L2(u)	Mean wake L2(omega)	Max OOS orth defect
FP16 V100	100/0	1000	9.43 h	1.887%	25.578%	3.318%	36.051%	0.469
FP16 V50/W50	50/50	650	3.90 h	1.945%	27.151%	3.512%	39.782%	1.400
FP64 V100	100/0	5000	115.82 h	1.821%	23.655%	3.435%	36.804%	0.457
FP16 W100	0/100	5000	29.43 h	1.708%	20.753%	3.109%	32.679%	0.756

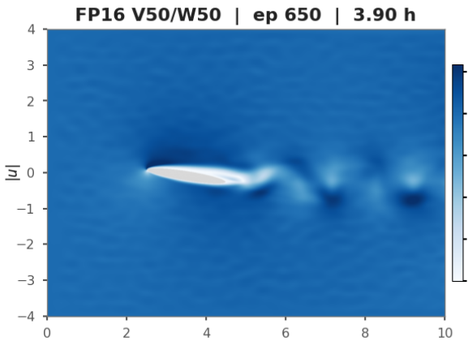
Lower bars indicate better reconstruction. All four rows use identical GT fields and velocity-L2 coefficient fitting; the table reports the observed out-of-sample orthogonality defect.

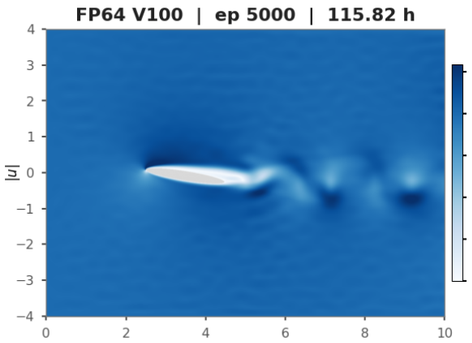
Upper frame: $\beta = 0.50$, $\alpha = +10^\circ$ | Re=1500, frame 20, velocity-L^2 coefficient fit

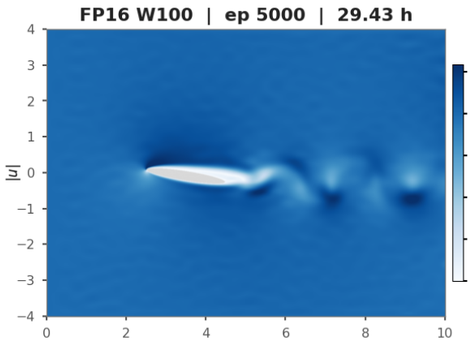
Same GT and plotting scales across all four models; full-domain and wake relative L2 errors are printed above each error map.

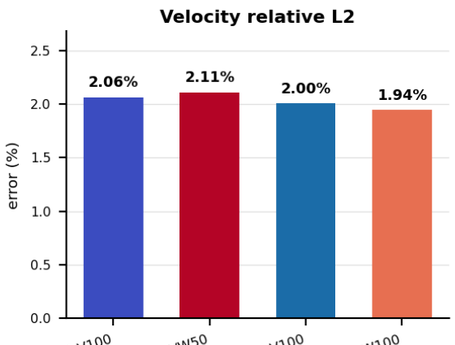


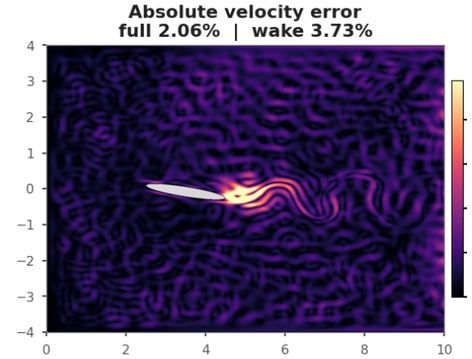


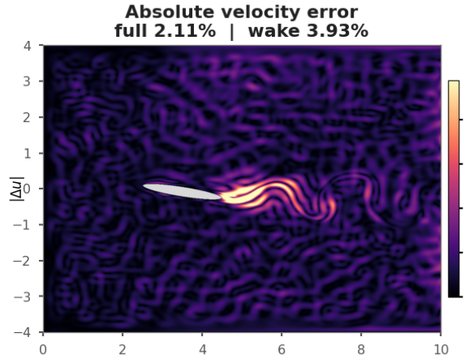


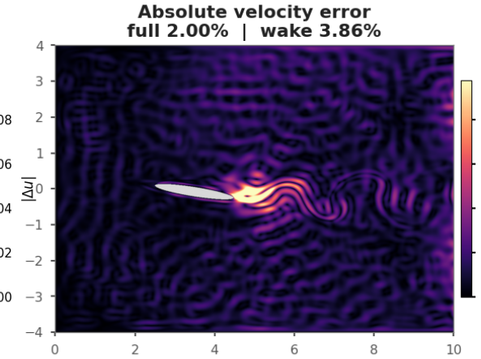


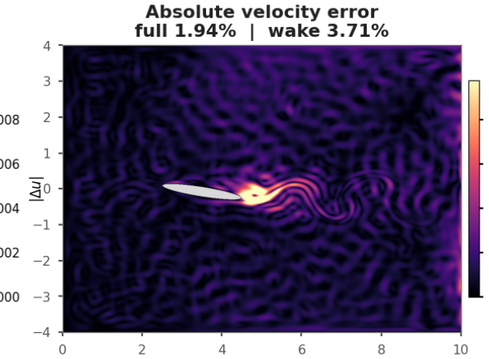


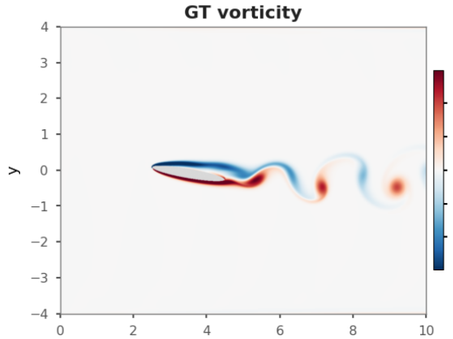


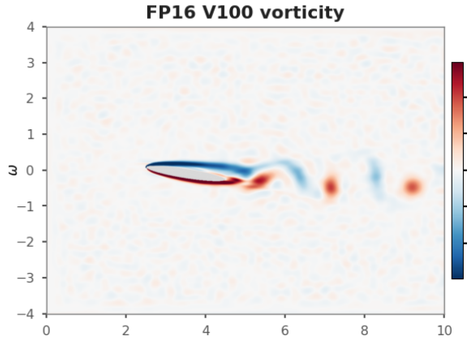


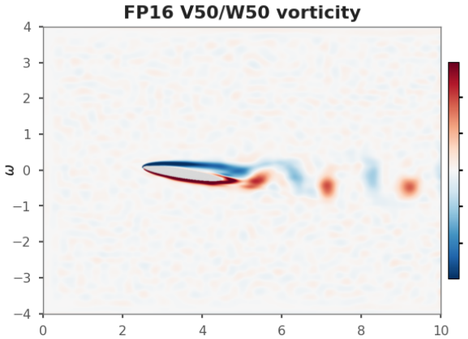


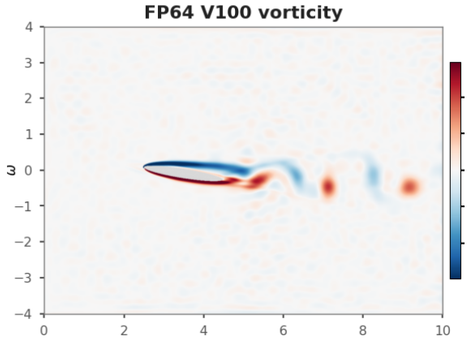


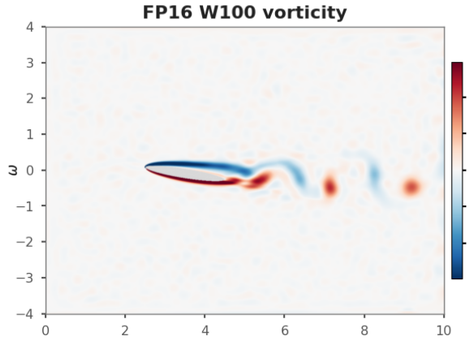


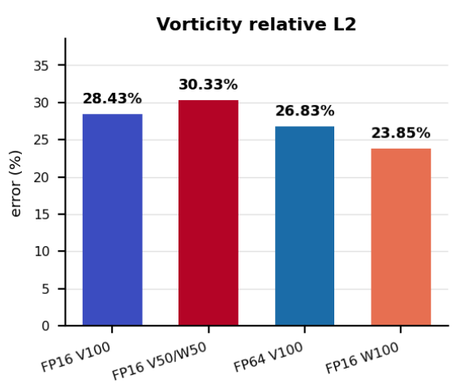


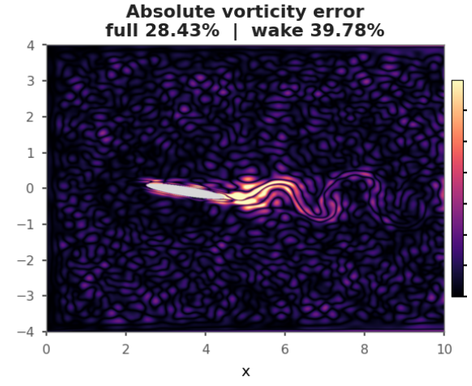


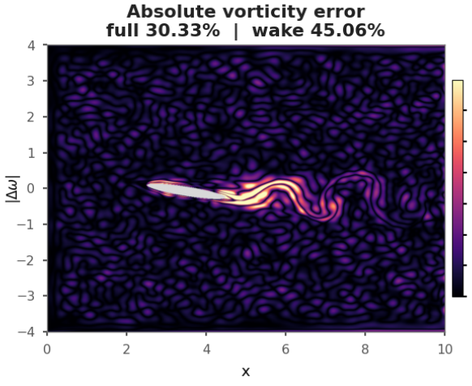


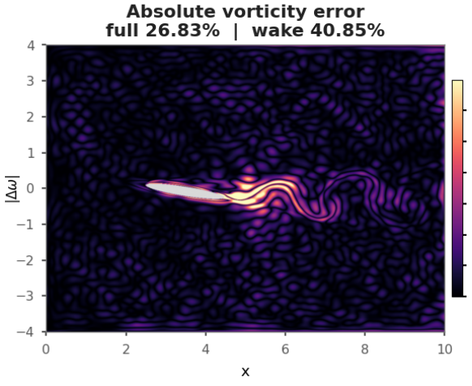


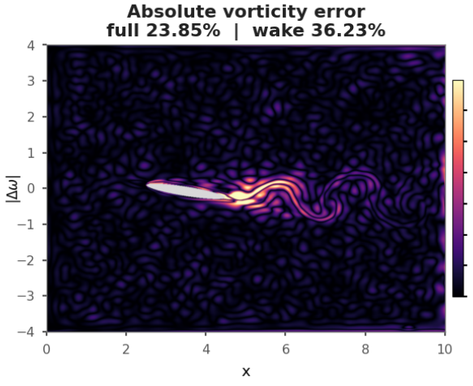








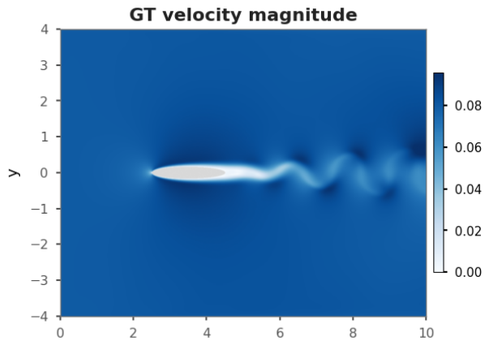


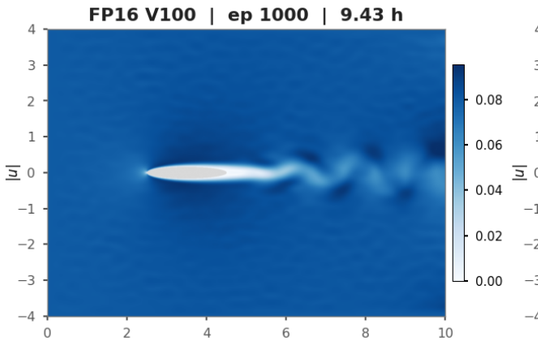


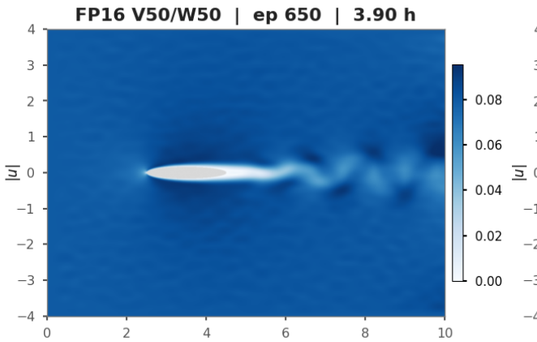
Gray region: solid airfoil. Error maps share one scale within each row. Metrics are evaluated on the full fluid grid; wake metrics use the fixed wake mask.

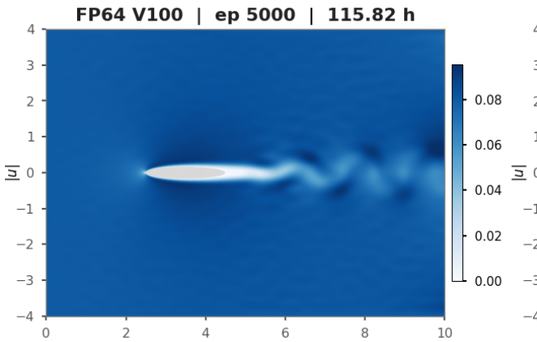
Middle frame: $\beta = 0.85$, $\alpha = 0^\circ$ | Re=1500, frame 20, velocity-L^2 coefficient fit

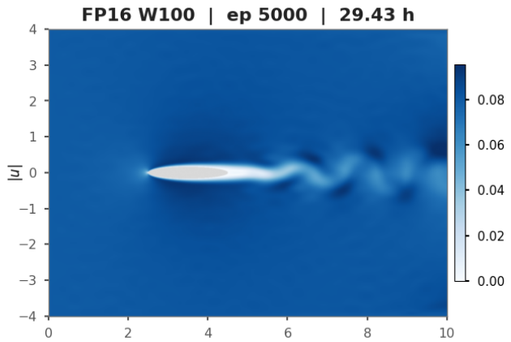
Same GT and plotting scales across all four models; full-domain and wake relative L2 errors are printed above each error map.

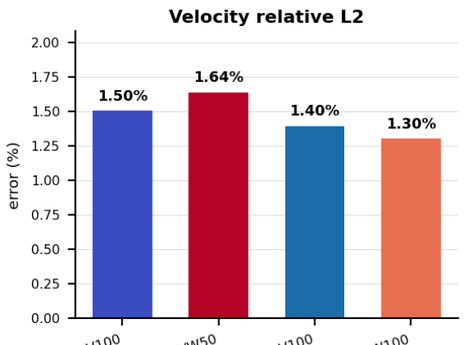


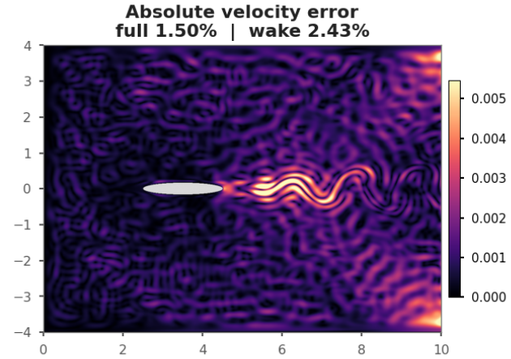


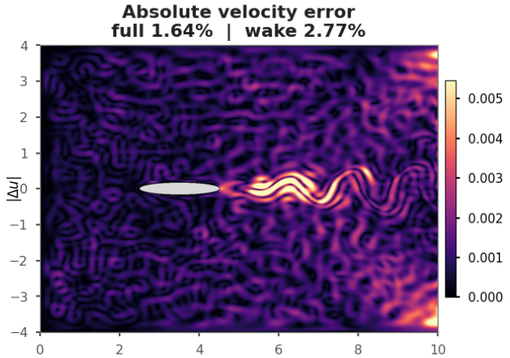


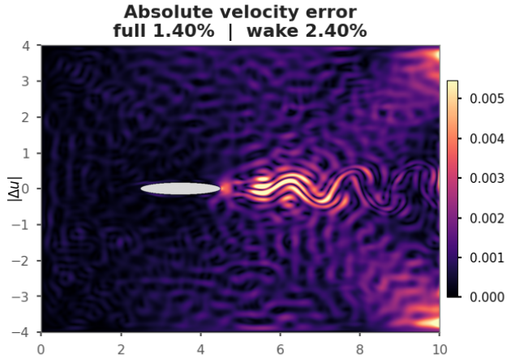


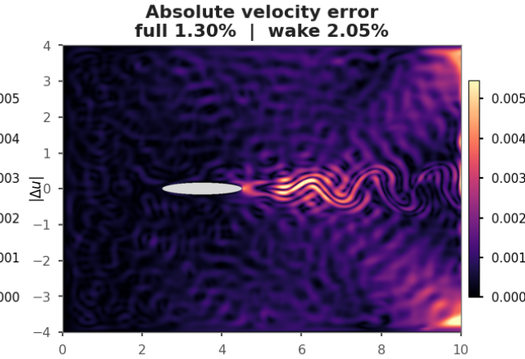


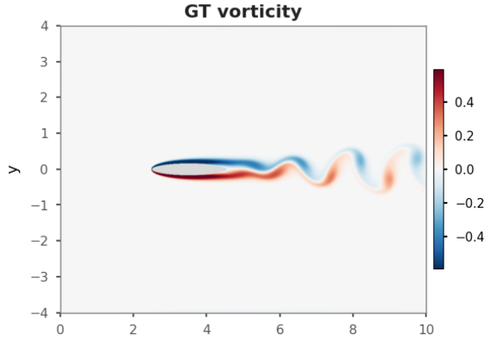


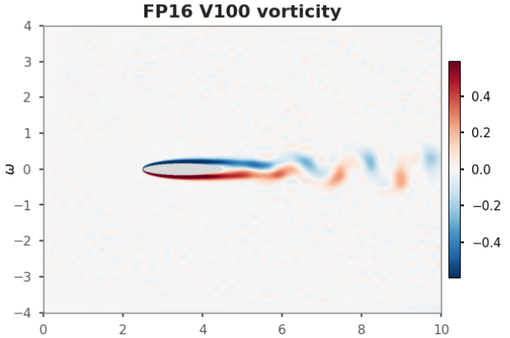


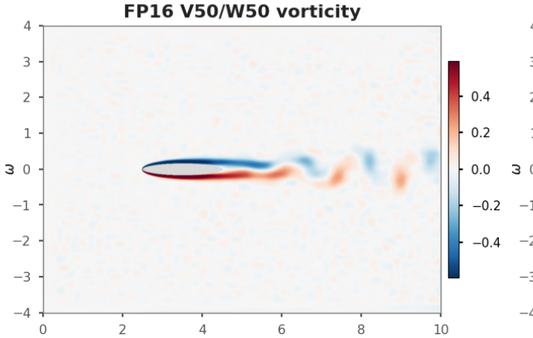


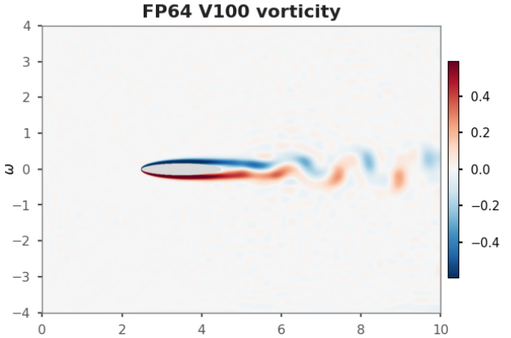


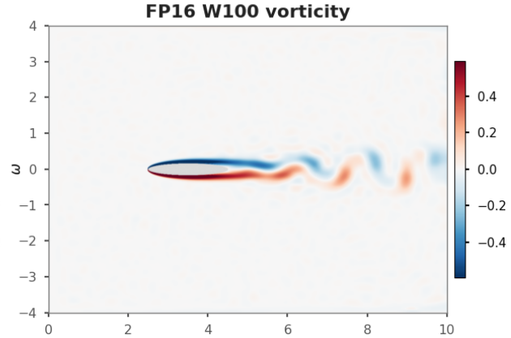


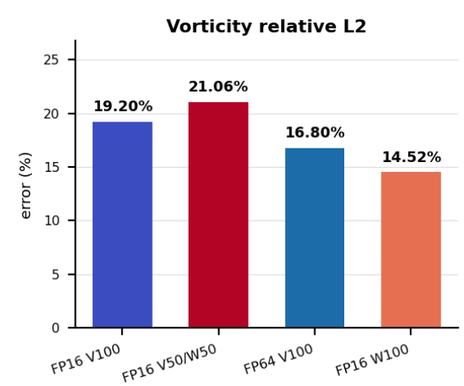


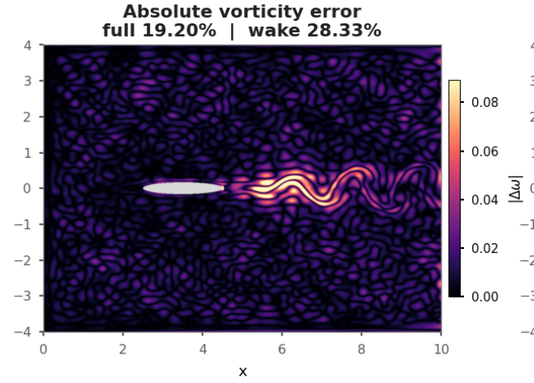


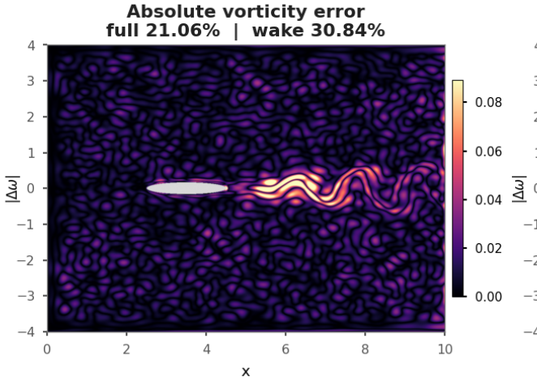


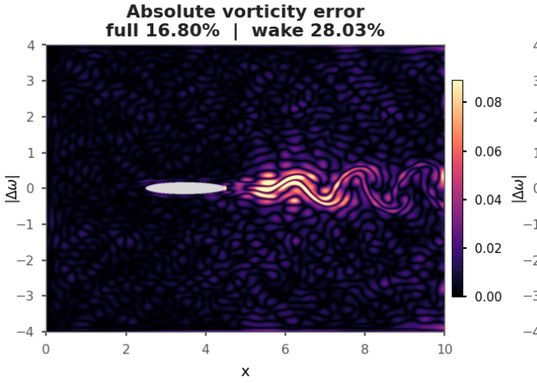


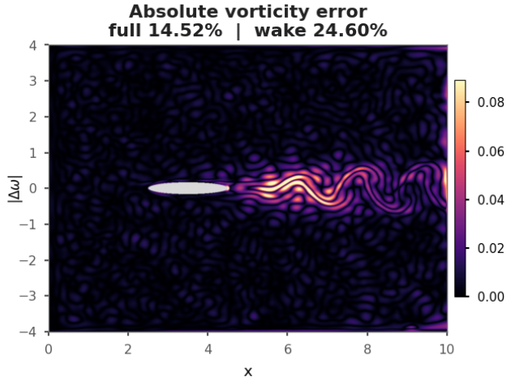








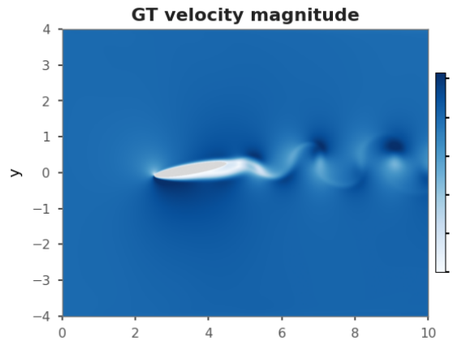


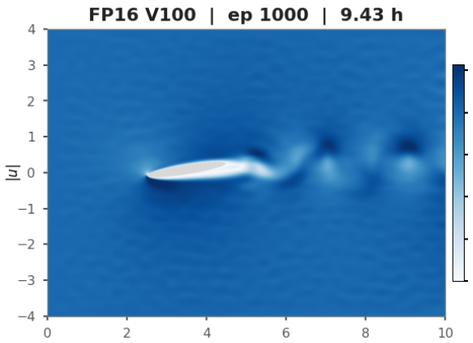


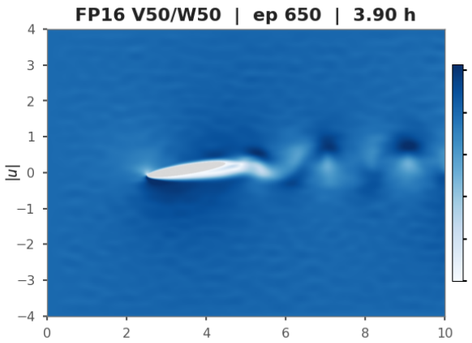
Gray region: solid airfoil. Error maps share one scale within each row. Metrics are evaluated on the full fluid grid; wake metrics use the fixed wake mask.

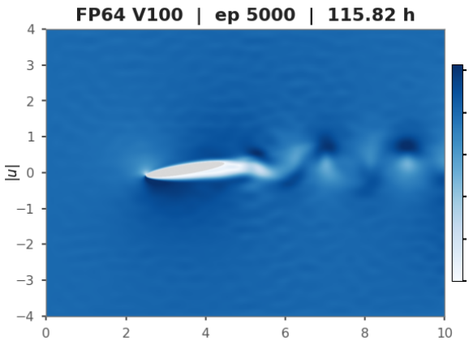
Lower frame: $\beta = 0.50$, $\alpha = -10^\circ$ | **Re=1500, frame 20, velocity-L^2 coefficient fit**

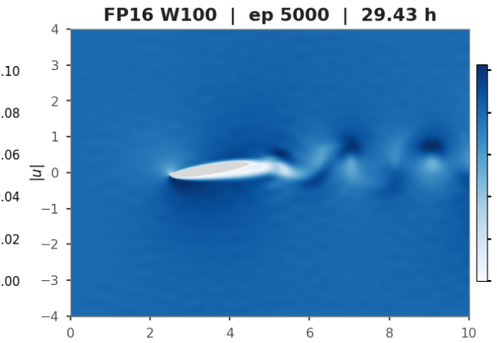
Same GT and plotting scales across all four models; full-domain and wake relative L2 errors are printed above each error map.

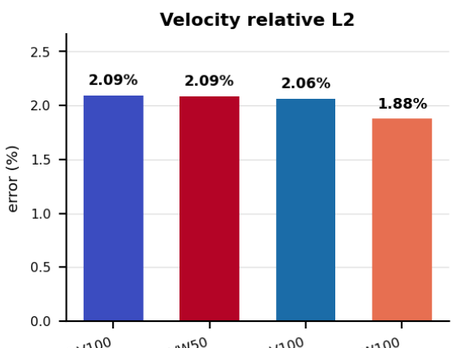


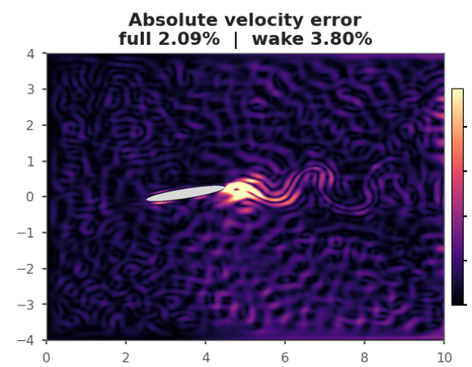


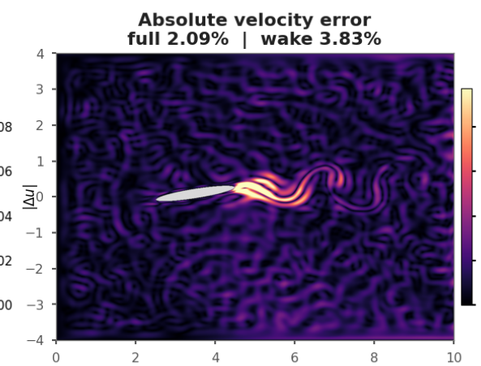


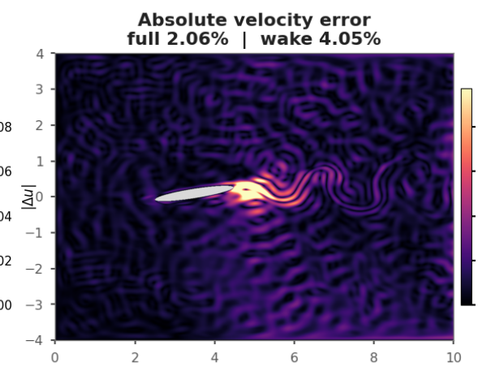


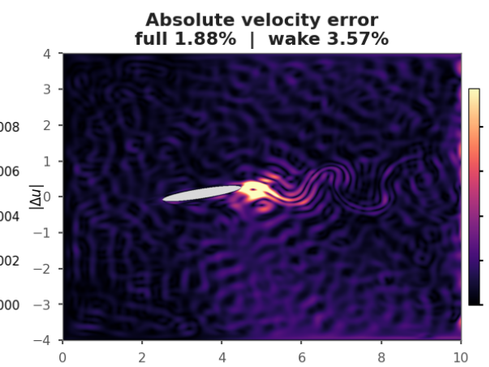


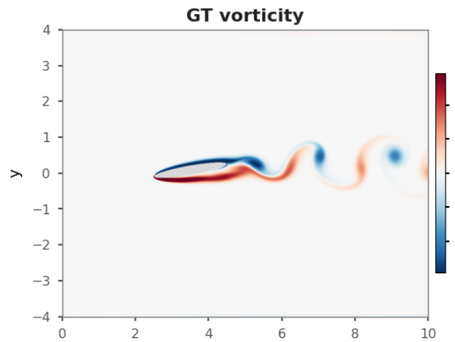


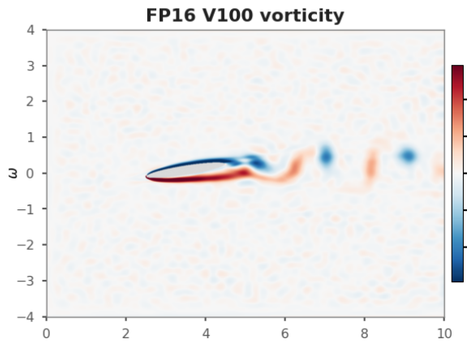


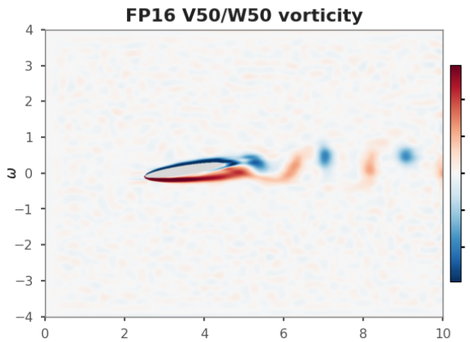


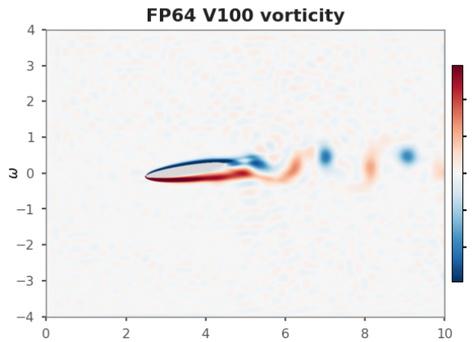


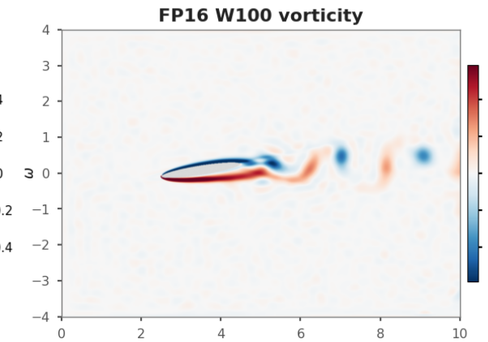


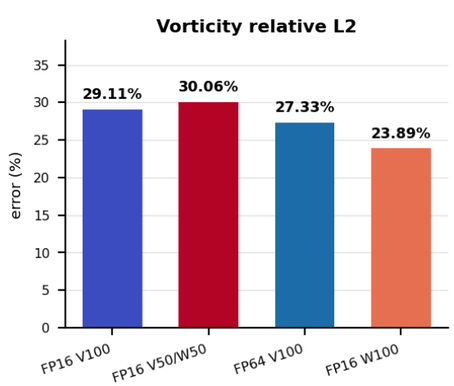


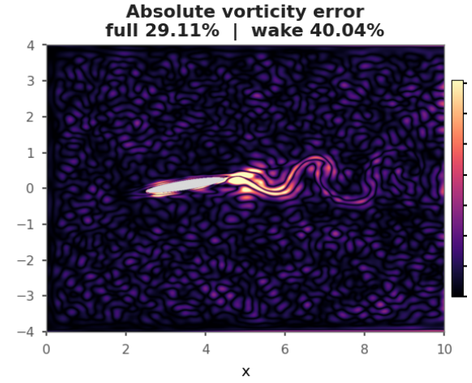


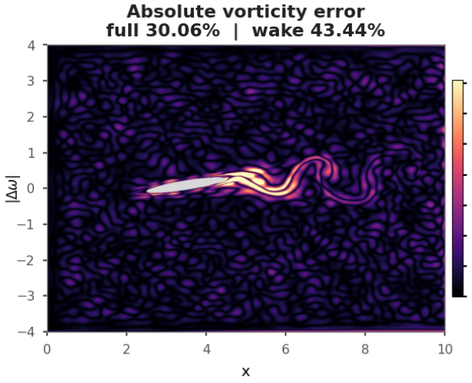


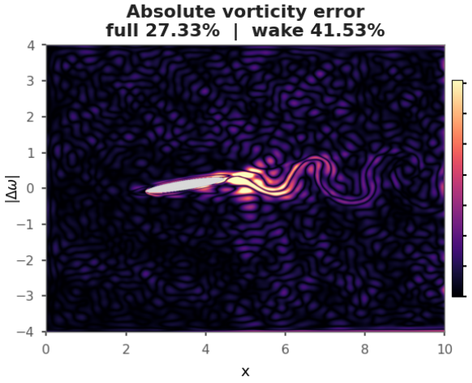


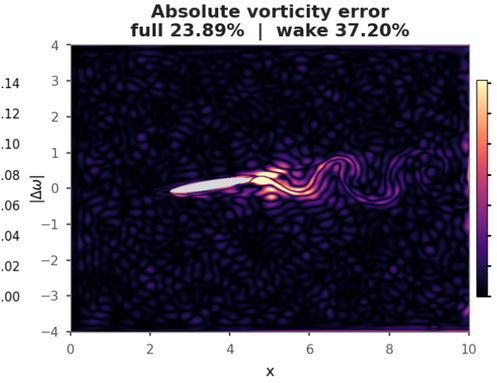












Gray region: solid airfoil. Error maps share one scale within each row. Metrics are evaluated on the full fluid grid; wake metrics use the fixed wake mask.